Recoil and explosions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use momentum conservation for an isolated system.

SUCCESS CRITERIA - I can...

1. Choose a system and check external impulse.
2. Add momentum with signs.
3. Explain equal-and-opposite momentum changes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “recoil” and “external” to explain your model or result. Sentence starters are optional.

recoil: motion backward when another part is sent forward

external: acting from outside the chosen system

conserved: unchanged in total under the stated conditions

01 Warm-up: bring back an earlier idea

A cart has momentum -15 kg·m/s and mass 3 kg.
Find velocity and its new momentum if velocity reverses at unchanged speed.

Word help: momentum = mass times velocity, including direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p_{total,\,i} = p_{total,\,f}$.
Negligible net external impulse. Internal forces can change each part’s momentum but not the system total.

Worked reasoning: Two carts initially at rest push apart with negligible external impulse. A 1 kg cart leaves at +4 m/s; a 2 kg cart must have momentum -4 kg·m/s, hence velocity -2 m/s.

Your turn: Draw before-and-after momentum arrows for the two-cart system.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two carts push apart from rest on an ideal track. **Draw** their momentum arrows. How must the two momenta compare?

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .



A moves left B moves right

Marzano check - circle: 1 2 3

04 Apply the physics

An astronaut of mass 60 kg and a 3 kg tool are initially at rest relative to a spacecraft. The astronaut pushes the tool so it moves at +8 m/s in that frame. **Label** the model and find the astronaut's velocity. Neglect external impulse.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Before **After**

Label objects and signed velocities.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Why can the two parts gain kinetic energy during a release while total momentum remains zero?

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A projectile's velocity is (12, -16) m/s. **Find** its speed. What information would locate it? **Label** separate horizontal and vertical motion.

Word help: projectile = an object moving after launch under the stated forces

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Two fragments of masses 2 and 3 kg separate from rest. The 2 kg fragment moves at +9 m/s.
Find the other velocity and total kinetic energy. Organize the known quantities on the before-and-after model.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 60 kg person and 10 kg board start at rest. After a push, the board moves at +3 m/s relative to the ground.
Find the person's ground velocity and the board's speed relative to the person. Organize the known quantities on the before-and-after model.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found ... [unit]. This value means ... in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3